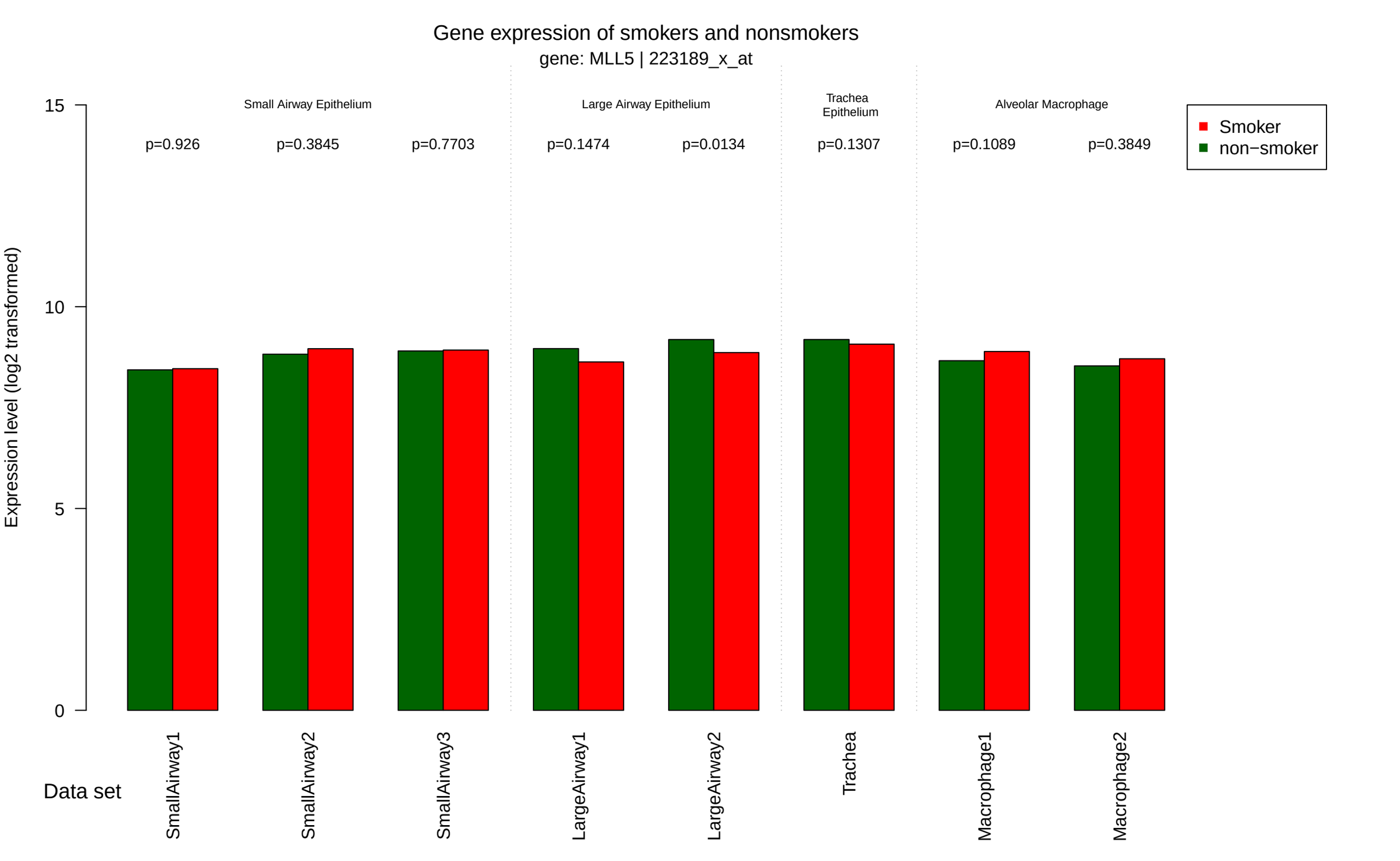
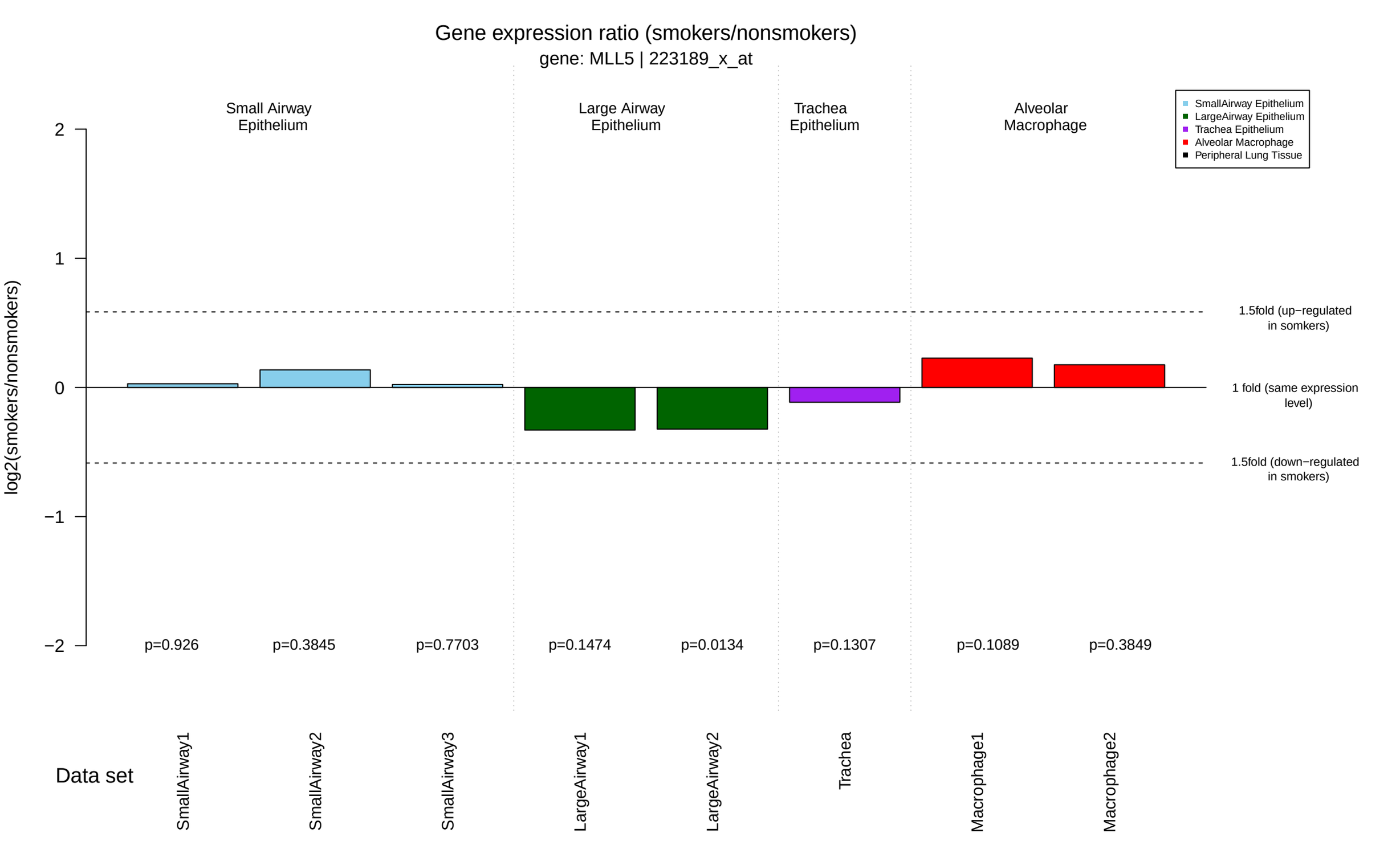
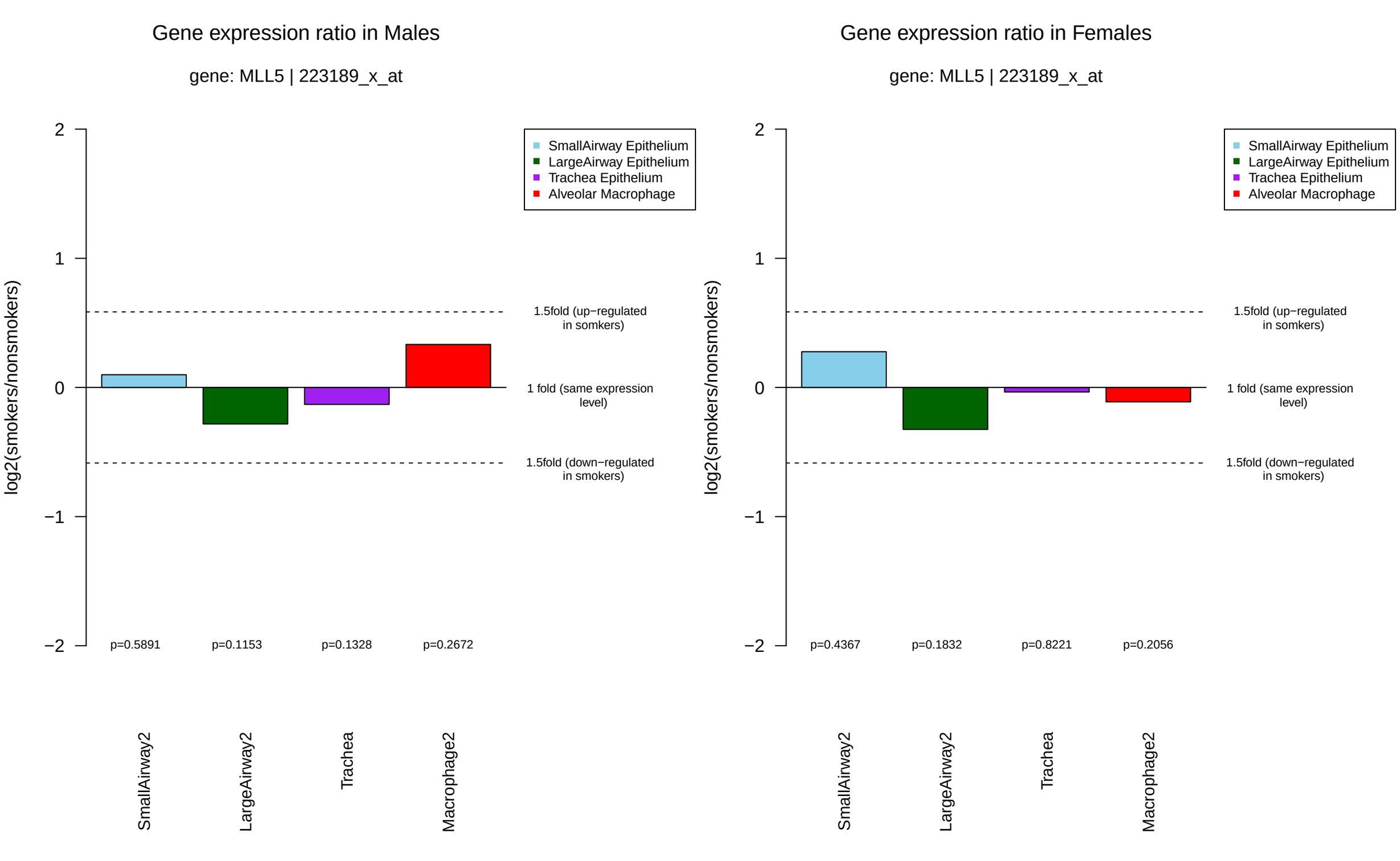
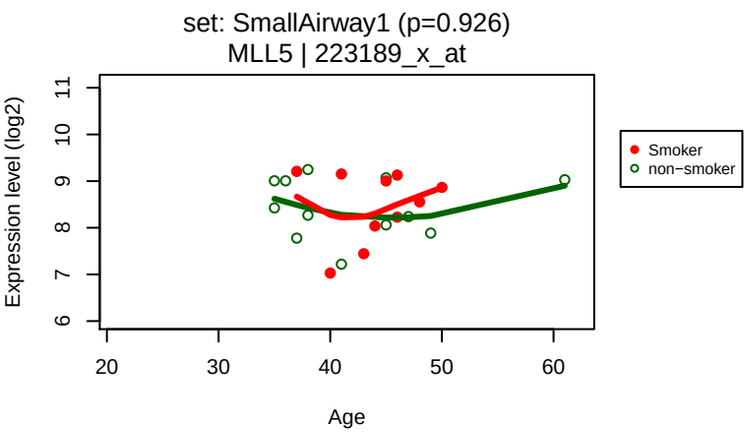
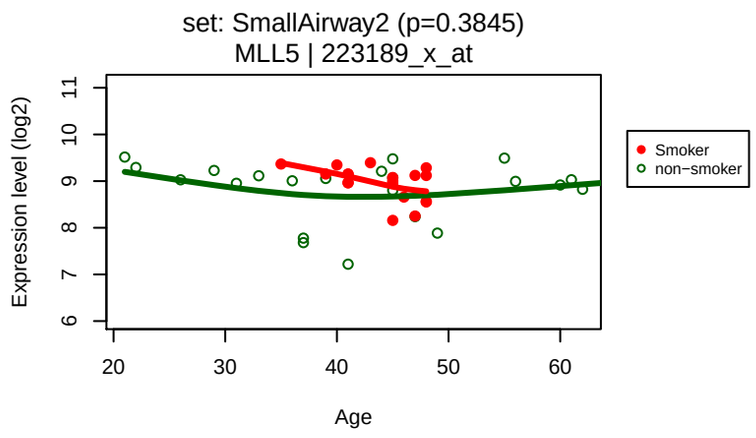
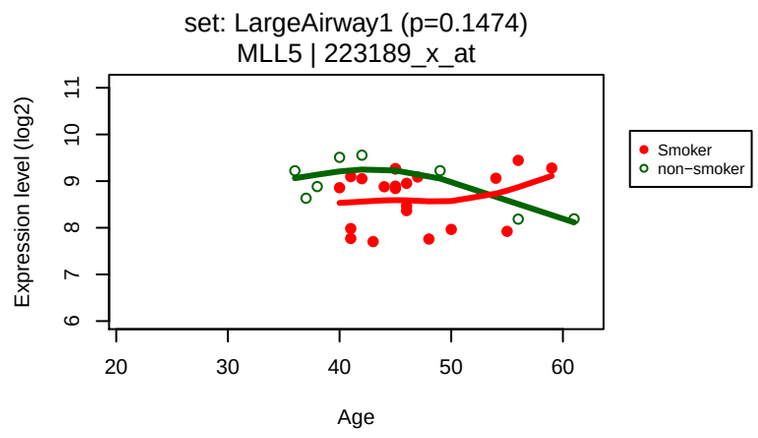

Gene expression of smokers and nonsmokers
gene: MLL5 | 223189_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.926
p=0.3845
p=0.7703
p=0.1474
p=0.0134
p=0.1307
p=0.1089
p=0.3849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

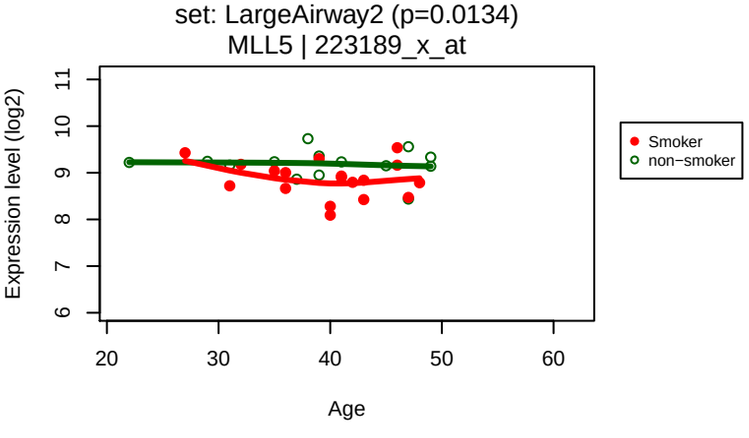

Gene expression ratio (smokers/nonsmokers)
gene: MLL5 | 223189_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.926
p=0.3845
p=0.7703
p=0.1474
p=0.0134
p=0.1307
p=0.1089
p=0.3849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

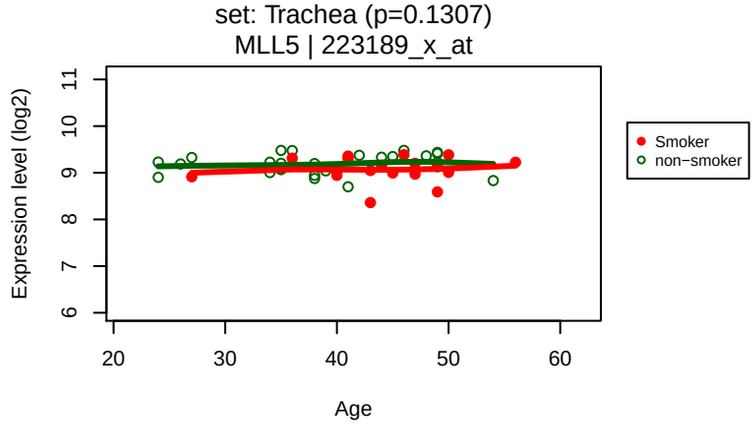

Gene expression ratio in Males
gene: MLL5 | 223189_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5891
p=0.1153
p=0.1328
p=0.2672
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MLL5 | 223189_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4367
p=0.1832
p=0.8221
p=0.2056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

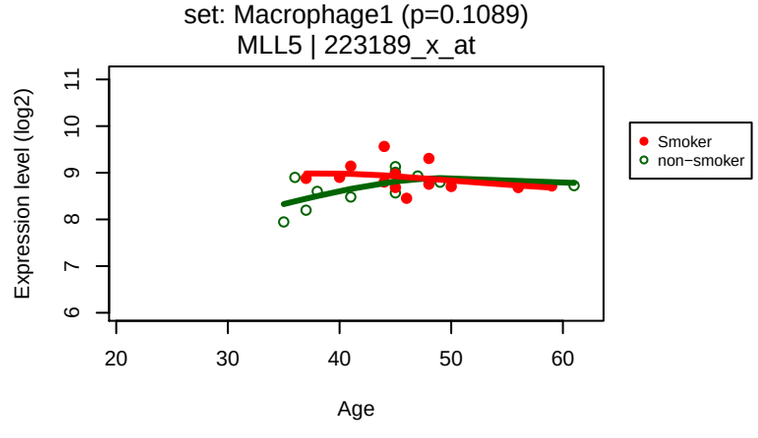

set: SmallAirway1 (p=0.926)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker

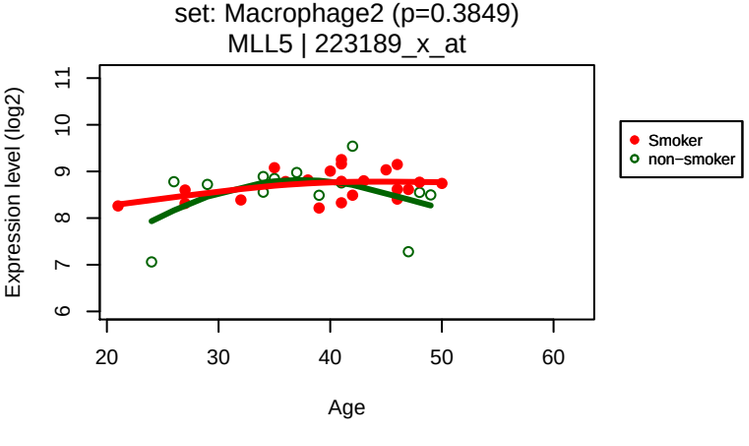

set: SmallAirway2 (p=0.3845)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker

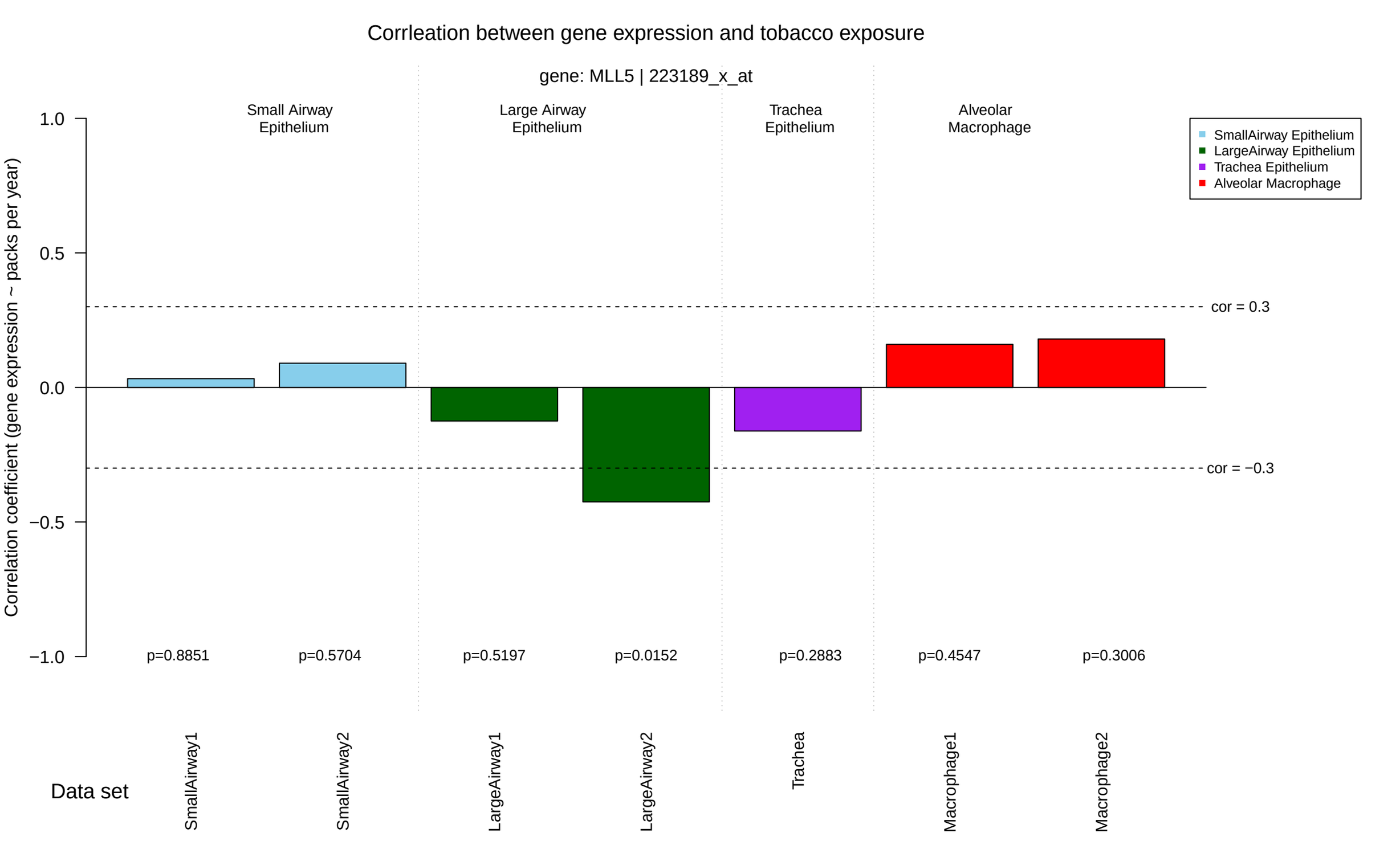

set: LargeAirway1 (p=0.1474)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0134)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1307)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1089)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3849)
MLL5 | 223189_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MLL5 | 223189_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8851
p=0.5704
p=0.5197
p=0.0152
p=0.2883
p=0.4547
p=0.3006
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2